Vulnerable Voxel Prediction and Monitoring for Semantic Map Integrity under Camera–LiDAR Misalignment

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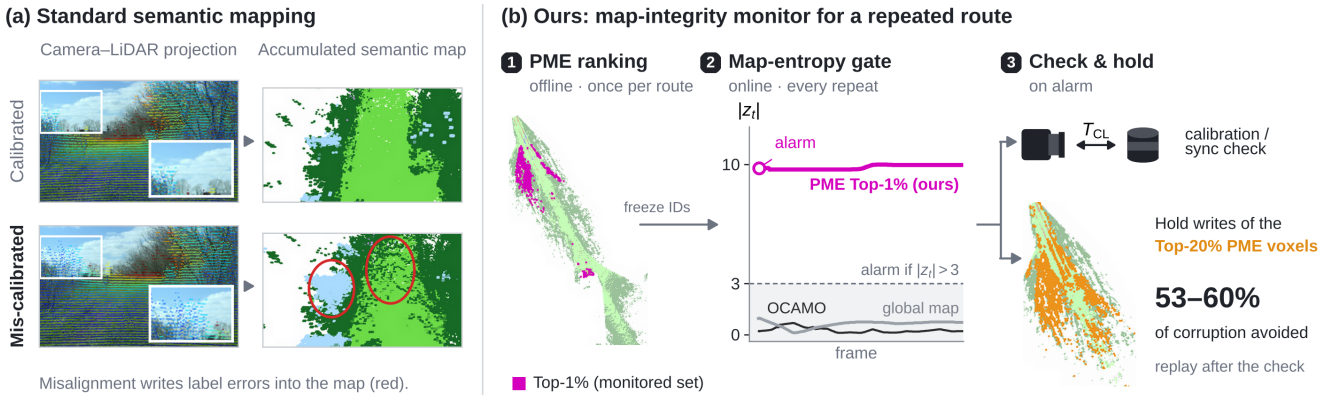


Fig. 1. **Map-integrity monitor for a repeated route.** (a) Miscalibration corrupts the accumulated map (red circles). (b) PME fixes the monitored set (Top-1%, magenta) once per route (1), a gate on its map entropy raises an alarm (2), and the robot then checks the calibration or synchronization and holds the writes of the protected set (Top-20%, orange) until the check (3).

***Abstract*— Camera–LiDAR semantic mapping writes every label through the extrinsic calibration, so drift or a time offset silently corrupts the map. Online calibrators and calibration monitors judge only the sensor pair, not the map, and a constant-extrinsic calibrator cannot remove a time offset. We observe that misalignment concentrates label errors in a few voxels whose projections cross label changes, which map-wide statistics dilute. For robots that repeat a route, such as patrol and inspection robots, we turn this observation into a training-free monitor that supervises rather than replaces calibration. A projection-marginalized entropy ranks vulnerable voxels from one clean pass of the route. A label-free gate on the map entropy of a small monitored set decides when to check the calibration, and the mapper holds the updates of a wider protected set until then. On urban and off-road data, including next-day re-drives, the gate detects 91% of the runs that lose over 5% mIoU, including runs a calibration monitor deems valid, almost never alarms without an offset, and flags a real image–scan pairing fault (a 137 ms time offset). Holding the protected set avoids 53–60% of the corruption.**

Index Terms—Sensor fusion, mapping, calibration and identification, semantic scene understanding, failure detection and recovery

I. Introduction

Many autonomous robots build 3D semantic maps with a camera–LiDAR pair. The mapper projects every LiDAR point into the camera image, reads the predicted label at that pixel, and accumulates the labels in a 3D voxel map, so the map is only as correct as the extrinsic calibration. Vibration and thermal change gradually shift this extrinsic [1]–[3], and online correction leaves residual drift between recalibrations. A camera–LiDAR time offset has the same effect while the robot moves. Every point then reads its label from a slightly wrong pixel, and the mapper silently writes the error into the map. This paper targets robots that repeat the same route, such as patrol and inspection robots.

Existing approaches act on the sensor pair, not on the map. Online calibrators re-estimate the extrinsic [1]–[3], and calibration monitors decide whether the pair remains calibrated [4]–[6]. Both model a constant T_{CL} and make one decision for the pair, which leaves three gaps. First, their validity signal does not tell whether the map is still correct. OCAMO's index stays valid on maps that have clearly lost accuracy (Sec. IV-E). Second, they cannot remove errors that a constant extrinsic does not describe, such as a camera–LiDAR time offset, which acts as an extrinsic error that changes with ego-motion. Third, when they run periodically, they keep re-estimating an extrinsic that is still correct and can disturb the map they should protect, and on RELLIS-3D no tested calibrator restores the map (Sec. IV-E). A global statistic such as the mean map entropy does not fill these gaps either, as averaging dilutes local damage.

Our key idea is to read drift from the few voxels where its damage concentrates. Voxels whose projection moves quickly under a small extrinsic perturbation and whose predicted label changes nearby absorb most label errors, while the rest of the map changes little (Fig. 1). We call this concentration *Localized Failure Amplification (LFA)* and explain it with a first-order projection-Jacobian analysis of a per-point *drift sensitivity*. The same voxels reveal drift without labels, because drift moves their projections across label changes and shifts their map entropy while the homogeneous interior barely changes.

We turn LFA into a lightweight two-stage map-integrity monitor that detects drift damage from the map itself (Fig. 2). It needs one clean commissioning traversal and reads the

same voxels on every repeat, so it does not apply to a route driven for the first time, where online calibration remains the tool. Offline, projection-marginalized predictive entropy (PME), the entropy of the mean label distribution over Monte Carlo perturbed projections, estimates the drift sensitivity from one clean commissioning traversal and fixes the Top-1% voxels as the monitored set. Online, a label-free gate tracks the map entropy of the monitored set and, when it leaves a commissioning baseline, holds the updates of the protected set, the wider Top-20% by PME, until a calibration or synchronization check. The gate estimates no extrinsic. It decides when the robot needs a calibrator and protects the map until the correction arrives.

On RELLIS-3D, KITTI-360, and our off-road data, including next-day re-drives, the gate detects 91% of the runs that lose over 5% mIoU, including runs OCAMO deems valid, almost never alarms without an offset, and flags a real image–scan pairing fault (a 137 ms time offset). Holding the protected set avoids 53–60% of the corruption.

This paper makes three contributions.

- We propose a label-free, training-free map-integrity monitor that decides when to check the calibration and what to hold, so that a robot on a repeated route keeps miscalibration out of its map.
- We identify LFA, the concentration of drift damage in a few voxels, on three datasets and three backends and explain it with a first-order projection-Jacobian analysis, which shows why map-wide statistics dilute it.
- We introduce PME, a training-free score that finds these voxels from one clean traversal and that boundary, depth, or uncertainty cues do not reproduce.

II. Related Work

Camera–LiDAR extrinsic calibration. Calibration methods range from target-based procedures [7] and information-theoretic alignment [8] to deep regression networks that estimate miscalibration during operation [1]–[3]. Levinson and Thrun [6] and the two methods OCAMO and LTO [4] monitor a reference calibration and track drift online, *Cal or No Cal?* classifies the calibration state of each frame as calibrated or miscalibrated [5], the concurrent MiCaDe [24] detects miscalibration changes online, and MIAS-LCEC [1] and the training-free Calib-Anything [9] re-estimate the extrinsic by matching LiDAR points to image segmentation masks. Motion-based methods compare camera and LiDAR ego-motion to detect and correct miscalibration [10]. All of them make one decision about T_{CL} for the whole sensor pair and do not act on the map. Our gate does not estimate T_{CL} and acts on the map, so it complements these methods.

Robustness and uncertainty. Corruption benchmarks for 3D detection include perturbed extrinsics [11] but report scene-level metrics. MC Dropout [12] and evidential heads [13] quantify model uncertainty. Predictive entropy [14], our uncertainty baseline, assumes a fixed projection.

3D semantic mapping and map health. Dense semantic mappers [15], [16] aggregate per-frame predictions into voxel or continuous maps, some with evidential weighting [16], and assume accurate registration during label transfer, which drift violates. Map-health monitoring spans localization integrity [17] and introspective perception [18]. Unlike these global summaries, we monitor the voxel subset that LFA singles out.

III. Methodology

A. Why Drift Damage Localizes

A first-order analysis shows which voxels drift damages and why this is not the same as boundary proximity. Let $\delta\boldsymbol{\xi} = [\delta\boldsymbol{\omega}^\top, \delta\mathbf{t}^\top]^\top$ be a small SE(3) perturbation with covariance Σ_ξ. For a 3D point $\mathbf{p}_k$ with clean camera-frame coordinate $\mathbf{c} = R\mathbf{p}_k + \mathbf{t}$ and clean projection $\mathbf{u}_0$, let $\mathbf{r}_k \equiv R\mathbf{p}_k$ denote the clean rotated point before translation. The linearized image-plane perturbation is

$$\delta\mathbf{u} \approx \mathbf{J}_{\pi,\xi}\,\delta\boldsymbol{\xi}, \quad \Sigma_u = \mathbf{J}_{\pi,\xi}\Sigma_\xi\mathbf{J}_{\pi,\xi}^\top, \tag{1}$$

where

$$\mathbf{J}_{\pi,\xi} = \frac{\partial\mathbf{u}}{\partial\mathbf{c}}\begin{bmatrix}-[\mathbf{r}_k]_\times & \mathbf{I}\end{bmatrix}, \quad \frac{\partial\mathbf{u}}{\partial\mathbf{c}} = \begin{pmatrix}\frac{f_x}{c_z} & 0 & -\frac{f_x c_x}{c_z^2}\\ 0 & \frac{f_y}{c_z} & -\frac{f_y c_y}{c_z^2}\end{pmatrix}. \tag{2}$$

Applying a rotation perturbation as $\exp([\delta\boldsymbol{\omega}]_\times)R$ with a fixed $\mathbf{t}$ moves the camera-frame point by $\delta\mathbf{c} \simeq -[\mathbf{r}_k]_\times\delta\boldsymbol{\omega}$, the rotational block of Eq. (2). Let $\mathbf{f}$ denote the per-pixel softmax map of the segmenter, $\Delta\mathbf{f}_k = \mathbf{f}(\mathbf{u}_0+\delta\mathbf{u}) - \mathbf{f}(\mathbf{u}_0)$ the change of the label distribution that point k reads, and $\Lambda_f \approx \mathbf{J}_f^\top\mathbf{J}_f$, where $\mathbf{J}_f$ is the spatial Jacobian of $\mathbf{f}$. The expected squared distribution change then becomes

$$\mathbb{E}\big[\|\Delta\mathbf{f}_k\|_2^2\big] \;\approx\; \mathrm{tr}(\Lambda_f(\mathbf{u}_0)\,\Sigma_u) = \mathrm{tr}\big(\mathbf{J}_{\pi,\xi}^\top\Lambda_f(\mathbf{u}_0)\mathbf{J}_{\pi,\xi}\Sigma_\xi\big)\,. \tag{3}$$

We call $s_k \equiv \mathbb{E}[\|\Delta\mathbf{f}_k\|_2^2]$ the *drift sensitivity* of point k, and Eq. (3) is the basis of LFA. For the rotation-only perturbations of our main experiments, the scale of $[\mathbf{r}_k]_\times$ cancels the $1/c_z$ factor in $\partial\mathbf{u}/\partial\mathbf{c}$ to first order. Away from camera–LiDAR lever-arm effects, $\|\delta\mathbf{u}\| \approx (f/c_z)\|\mathbf{r}_k \times \delta\boldsymbol{\omega}\|$ for a representative focal length f stays approximately independent of depth and grows mainly with image eccentricity. Inverse-depth sensitivity instead enters through the translation block and the fixed lever arm $\mathbf{t}$, so a translation moves distant points less than a rotation does. Eq. (3) couples this geometric factor with Λ_f, which is nonzero only where the label field varies, so damage concentrates at semantic boundaries (often occlusion boundaries, where depth and label edges align) and map-wide averages dilute it. Sec. IV-C tests that their product ranks damage better than either factor.

B. Semantic Mapping and Evaluation Groups

We use a DeepLabV3 segmenter with a MobileNetV3-Large backbone [19], [20] and an evidential output head [13] for the controlled mapping and ranking experiments, because it supplies both softmax probabilities and the evidential parameters that the evidential mapper EBS [16] requires. We quantize space into a 0.3 m voxel grid and evaluate three backends. **S-CSM** [15] (semantic counting sensor model)

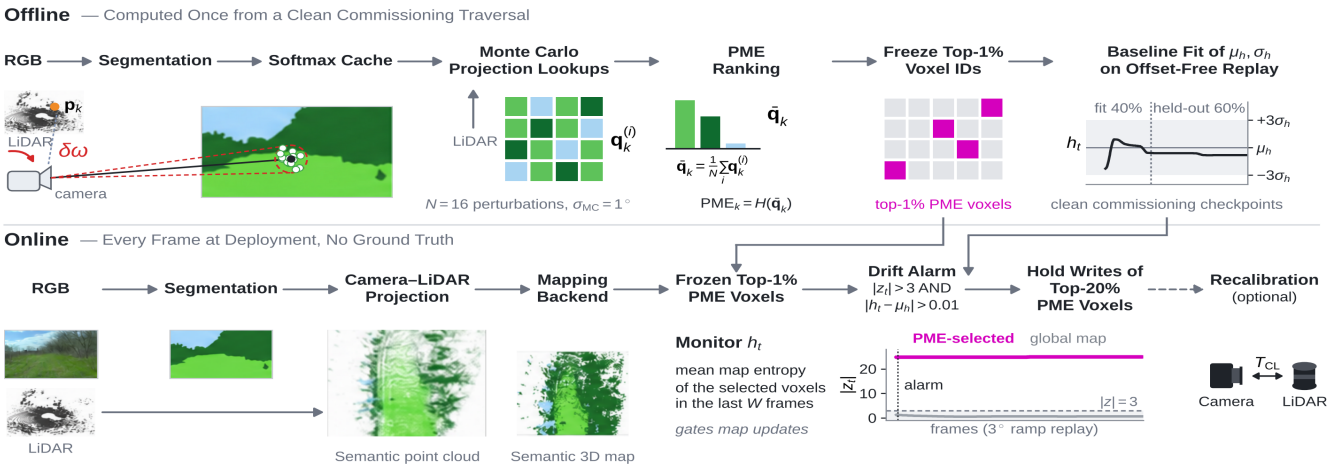


Fig. 2. **Map-integrity monitor pipeline.** Offline, PME ranks voxels by Monte Carlo projection lookups on one clean commissioning traversal and freezes the Top-1% set with the baseline μ_h, σ_h of its map entropy h_t. Online, the gate compares h_t with μ_h, σ_h, and an alarm holds the writes of the protected set (Top-20% PME voxels) until a recalibration (Sec. III-D).

averages per-point softmax scores uniformly, **S-BKI** [15] (semantic Bayesian kernel inference) weights each observation by a sparse spatial cosine kernel of its distance to the voxel center (ℓ=0.2 m), and **EBS** [16] additionally downweights each observation by the evidential vacuity $\nu_k \in [0, 1]$. Only EBS uses the vacuity, and all scores below use the softmax to stay comparable across backends.

To measure LFA, we compare the **Top-1%** voxels of a ranking score with the **Global** voxel set. The relative accuracy drop for group G is

$$\Delta_G^{(\sigma)} = \frac{\mathrm{mIoU}_G^{(0)} - \mathrm{mIoU}_G^{(\sigma)}}{\mathrm{mIoU}_G^{(0)}} \times 100 \quad [\%], \tag{4}$$

and the amplification ratio $\rho = \Delta_{\mathrm{Top1\%}}/\Delta_{\mathrm{Global}}$ divides the drop of the Top-1% group by that of the Global set. We compute mIoU on occupied, labeled voxels after label remapping, as the macro-average over classes with nonzero support (excluding void/unknown).

C. Offline Selection of Vulnerable Voxels

The first stage needs a score that estimates the drift sensitivity s_k of Eq. (3) without ground truth. We estimate it by Monte Carlo sampling instead of computing $\mathbf{J}_f$. We sample N SE(3) perturbations, $\delta\boldsymbol{\omega}_i \sim \mathcal{N}(\mathbf{0}, \sigma_{\mathrm{MC}}^2\mathbf{I})$ and $\delta\mathbf{t}_i \sim \mathcal{N}(\mathbf{0}, \sigma_t^2\mathbf{I})$, and apply each one as $R_i = \exp([\delta\boldsymbol{\omega}_i]_\times)\, R_{\mathrm{clean}}$ and $\mathbf{t}_i = \mathbf{t}_{\mathrm{clean}} + \delta\mathbf{t}_i$, with $\sigma_{\mathrm{MC}} = 1°$, $\sigma_t = 2$ cm, and $N = 16$ perturbations that we draw once with a fixed seed and share across all frames. One segmentation pass per image caches the softmax map. For point $\mathbf{p}_k$, $\mathbf{q}_k^{(i)} \in [0, 1]^C$ is the cached softmax vector that we sample bilinearly at its projection under $(R_i, \mathbf{t}_i)$, and $\bar{\mathbf{q}}_k = N^{-1}\sum_{i=1}^{N} \mathbf{q}_k^{(i)}$, so the N samples require only new projections. We define projection-marginalized predictive entropy (PME) as

$$\mathrm{PME}_k = H(\bar{\mathbf{q}}_k) = -\sum_{j=1}^{C} \bar{q}_{k,j} \log \bar{q}_{k,j}. \tag{5}$$

PME is high where a point's label is already ambiguous or where small perturbations move its projection onto different labels, and it decomposes exactly as $\mathrm{PME}_k = N^{-1}\sum_i H(\mathbf{q}_k^{(i)}) + \mathrm{JS}_k^{\mathrm{proj}}$. The first term measures local predictive ambiguity, and the equal-weight generalized Jensen–Shannon divergence

$$\mathrm{JS}_k^{\mathrm{proj}} = H(\bar{\mathbf{q}}_k) - \frac{1}{N}\sum_{i=1}^{N} H(\mathbf{q}_k^{(i)}) \tag{6}$$

measures perturbation-induced disagreement and relates most directly to Eq. (3). We average the per-point PME within each voxel, rank the voxels of the commissioning map, and freeze the Top-1% set as the monitored set.

D. Online Label-Free Gate

Signal. For voxel v, $\bar{\mathbf{f}}_v$ averages the softmax $\mathbf{f}(\mathbf{u}_k)$ that its points $k \in \mathcal{P}_v$ read at their current projections $\mathbf{u}_k$, and its map entropy is $H_v = -\sum_j \bar{f}_{v,j} \log \bar{f}_{v,j}$. The monitor tracks h_t, the count-stratified mean of H_v over the monitored set in the map that the most recent W frames accumulate. We split the voxels into four strata by the number of points they receive, take the stratum edges from the quartiles of this count on the first 40% of the commissioning replay, and skip strata with fewer than five voxels. We then average the four stratum means with equal weight, so that the growing point count of a filling map does not move the signal. Once the window fills, the monitor decides at the fixed dataset cadence that Sec. IV-A reports. Voxels of the monitored set that the window has not yet observed do not contribute, and a checkpoint that observes none of them makes no decision. We set W per dataset from clean data, as point density and run length differ (W=200/180/40 for RELLIS-3D/KITTI-360/off-road).

Baseline and alarm. An offset-free commissioning replay provides the baseline. We split the clean replay by frame index into a baseline segment, the first 40% of frames, and a disjoint clean false-alarm segment, the final 60%. We fit the mean μ_h and standard deviation σ_h of h_t using the eligible checkpoints in the baseline segment after the W-frame window has filled, and count clean false alarms only



Fig. 3. Off-road platform, (a) vehicle and (b) roof-mounted sensor rig.

at eligible checkpoints in the final 60%. We evaluate drifted replays over all eligible checkpoints from the first decision after the window fills. The alarm fires when both $|z_t|>3$, with $z_t = (h_t - \mu_h)/\sigma_h$, and $|h_t - \mu_h|>0.01$ hold, where the margin suppresses tiny deviations. All alarms use this full gate, without labels or a paired clean replay.

Action. After an alarm, the mapper buffers incoming frames and holds the writes of the protected set, the top-r voxels by commissioning PME (r=20%), until an external calibration or time-synchronization check corrects or confirms the camera–LiDAR alignment, and then replays the buffer (Sec. IV-E). A false alarm only delays updates, never damaging the map.

IV. Experiments

A. Experimental Setup

Datasets. RELLIS-3D [22] is a five-sequence off-road benchmark (Ouster OS1-64, 1920×1200 camera). **KITTI-360** [23] is an urban driving dataset (Velodyne HDL-64E), for which we use the front-view camera and sequences 03–06. We have collected **our off-road dataset** in unpaved environments with the vehicle-mounted sensor rig in Fig. 3, which carries a Blackfly 23S6C-C RGB camera, a Velodyne Puck 32MR 3D LiDAR, and an Applanix LV-210 GPS/IMU. It contains manually annotated images from five sequences, OF1–OF5, with four classes (unpaved, grass, vegetation, structure). For cross-sequence ranking, we hold out each sequence once (5/4/5 folds) and train and select on the others.

We fix all hyperparameters (σ_{MC}, N, Top-k, voxel size) and the complete monitoring protocol of Sec. III-D on RELLIS-3D, our development set, before we evaluate any KITTI-360 or off-road fold. During development, we have compared fixed z-score, k-of-n, CUSUM, and ratio alarms on RELLIS-3D and report only the retained rule.

Stronger-frontend audit. For each RELLIS-3D sequence, we fine-tune an ADE20K-pretrained SegFormer-B2 [21] on the other four, recompute PME and the monitored set from its held-out softmax, and replay the unchanged gate. Its held-out clean mIoU is 64.7±9.4%, against 36–54% for MobileNetV3–DeepLabV3.

Perturbation protocol. As the true extrinsic of a drifting platform is rarely measurable, we follow prior work [3], [5], [10] and inject controlled miscalibration into recordings, perturbing only the projection extrinsic within 0°–4°. (i) For mapping and ranking (Secs. IV-B and IV-C), each replay draws one $\delta\boldsymbol{\omega} \sim \mathcal{N}(\mathbf{0}, \sigma^2\mathbf{I})$ with per-axis $\sigma \in \{0.5°, 1°, 2°, 4°\}$ and applies it to all frames, so each replay sees a constant offset. We average over K=5 draws per fold, and only the AR(1) fault of Sec. IV-B varies across frames. (ii) For monitoring (Secs. IV-D and IV-E), which uses the S-CSM map (0.5 m voxels off-road), a **fixed offset** applies a constant rotation of magnitude θ from the first frame, and a **fast-then-held ramp** raises it linearly from 0° to θ over the first 15% of frames and holds it. Slow drift from vibration and thermal dilation [4], [24] is nearly constant within one run, so the fixed offset models it. Shocks, minor accidents, and rough handling [4], [25] shift the extrinsic within a fraction of the run, and the shift persists, so the ramp models them. A fixed offset also stands for any calibrator's residual after convergence, so monitoring results characterize the gate independently of the calibrator.

Comparison methods. Seg-Entropy [14] serves as the semantic-uncertainty ranking baseline. For monitoring, the identical gate watches the Top-1% sets of Seg-Entropy and of the semantic-boundary score of Fig. 5, a random 1% set, or the global map, and we run OCAMO [4] and LTO [4] from the authors' released code. We train the classifier of *Cal or No Cal?* [5] with the authors' code and optimizer (50+50 epochs) on the first 60% of each test sequence and test it on the last 40% (‡ in Table IV). In Table IV, Other domain trains the classifier on KITTI-360 03 without per-route retraining, – marks a test segment too short for a windowed mean, Setup is the compute before deployment (GPU training or our CPU clean pass of map update and PME), and ms is the per-frame cost (* A100 runtime from [5]).

Evaluation protocol. Every method passes its own score (V for OCAMO/LTO, P for the classifier) through the commissioning z-gate of Sec. III-D, either *per frame* or *windowed* on its mean over the last W frames, with decision strides of 20/5/5 frames, and the first 40% of each replay (of the test segment for the classifier) supplies μ_h, σ_h. The *native* rule is each method's own per-frame threshold. A run alarms if its rule fires at any decision, and run-level detection is the main metric. A *false alarm* is an offset-free run that alarms in its final 60%, and a *detection* is a drifted run that alarms anywhere. Because the gate detects change rather than damage, we also count detections among runs that lose over 5% mIoU of the final map, the loss at which we treat the map as needing recalibration. *Checkpoint AUROC* (Table IV) ranks the decision checkpoints in the final 60% of drifted replays against those of the offset-free replay by each raw statistic without the gate ($|h_t - \mu_h|$ for ours, $1-V$ after a 10-frame burn-in for OCAMO/LTO, P for the classifier), so it measures persistence and can be low although every run alarms. *Voxel AUROC* (Table II) ranks the voxels of one map against their realized damage and involves no alarm.

B. Localization of Drift Damage

Across the σ sweep of Fig. 4, the monitored set loses more relative accuracy than the global map for every $\sigma \geq 1°$, dataset, and backend. At the 1° operating point (Table I), ρ reaches 1.68–2.01× on RELLIS-3D and 1.86–2.65× on KITTI-360. The held-out SegFormer-B2 frontend raises it

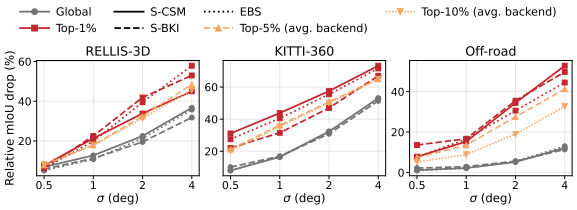


Fig. 4. Relative mIoU drop versus σ for Global and PME Top-1% by dataset and backend. Orange curves show the backend-averaged Top-5/10% groups.

TABLE I

RELATIVE MIOU DROP (%) OF THE TOP-1% AND GLOBAL VOXELS AT σ=1°. GRAY MARKS ρ THAT A SMALL GLOBAL DROP INFLATES.

Dataset	Map	Global	Top-1%	Δ_{pp}	ρ
RELLIS-3D	S-CSM	12.7	21.3	8.6	1.68×
	S-BKI	11.2	22.5	11.3	2.01×
	EBS	10.8	19.8	9.0	1.83×
KITTI-360	S-CSM	16.5	43.8	27.3	2.65×
	S-BKI	16.9	31.5	14.6	1.86×
	EBS	16.4	40.5	24.1	2.47×
Off-road	S-CSM	2.2	15.3	13.1	6.95×
	S-BKI	2.9	16.7	13.8	5.76×
	EBS	2.2	16.5	14.3	7.50×

to 2.54× on RELLIS-3D, so a stronger segmenter does not weaken LFA. Off-road Global drops are small and inflate ρ, and there the monitored set still loses 13.1–14.3 points more than the whole map (Δ_{pp} in Table I).

Structured faults. Single-axis, axis-coupled, and AR(1) faults also amplify the damage at 1° (ρ=2.04–3.12× on RELLIS-3D, 1.90–2.41× on KITTI-360).

LFA in the label-free signal. Without labels, the mean map-entropy change of the monitored set exceeds that of the global map across the Gaussian sweep by 11.8–30.7× on RELLIS-3D, 9.4–29.9× on KITTI-360, and 1.6–6.5× off-road, against 1.0–1.2× for Random 1% sets.

C. Selection of Vulnerable Voxels

We define voxel-level damage as the realized relabeling rate, the fraction of repeated σ=4° all-frames-perturbed rebuilds in which the predicted label of a voxel differs from its clean label, and label the upper 10% damage quantile, including ties at the cutoff, as positive (21.9, 34.8, and 12.8% of the voxels on RELLIS-3D, KITTI-360, and off-road).

PME attains the highest AUROC on all three datasets and the highest AUPRC on KITTI-360 and off-road. Its disagreement component JS^{proj} alone matches PME on RELLIS-3D but falls below matched random in tail precision off-road (P@1 0.401 vs. 0.631). Simple voxel attributes do not explain the gain. PME improves Top-1% precision over the all-matched random baseline, which controls boundary, depth, eccentricity, density, and class, on all 14 held-out splits (5/4/5 folds, one-sided Wilcoxon signed-rank $p<0.001$, 95% CI of the paired gain above zero).

In Fig. 5, random, boundary, and depth-discontinuity scores select many cells far from the damage, whereas PME concentrates near strong semantic transitions. We test simpler explanations of the selection result on RELLIS-3D

TABLE II

BACKEND-INDEPENDENT VOXEL DAMAGE RANKING AGAINST σ=4° REBUILDS. P@1 IS THE TOP-1% PRECISION.

Dataset	Score	Voxel AUROC	AUPRC	Spearman	P@1
RELLIS	PME	**0.793**	0.489	0.526	0.637
	JS^{proj}	0.791	**0.501**	**0.531**	**0.775**
	Seg-Entropy	0.706	0.391	0.350	0.543
	all-matched random	—	—	—	0.566
KITTI-360	PME	**0.755**	**0.600**	**0.420**	**0.810**
	JS^{proj}	0.736	0.556	0.390	0.699
	Seg-Entropy	0.725	0.562	0.372	0.729
	all-matched random	—	—	—	0.717
Off-road	PME	**0.908**	**0.577**	0.566	**0.753**
	JS^{proj}	0.879	0.410	**0.570**	0.401
	Seg-Entropy	0.880	0.524	0.491	0.658
	all-matched random	—	—	—	0.631

TABLE III

GATE ALARMS ON THE FIXED-OFFSET RUNS IN BINS OF FINAL-MAP MIOU LOSS (%). FIRST ALARM IS IN % OF THE RUN, AND PREVENTED IS THE SHARE OF NEW LABEL ERRORS AVOIDED. OF THE 84 FIXED-OFFSET RUNS, 78 HAVE A FINAL DAMAGE MAP (SIX KITTI-360 RUNS LACK ONE). THE 5% TOL. ROW SPLITS THEM AT A 5% LOSS, AND 10 OF THE 12 ALARMS BELOW IT ARE OFF-ROAD.

	Clean	<2	2–5	5–10	10–20	>20
All runs (78 drifted, 14 offset-free)						
Runs	14	8	13	16	20	21
PME gate	0/14	3/8	9/13	14/16	18/20	20/21
PME gate, 5% tol.	–	12/21			52/57	
First alarm (%)	–	44	49	31	29	21
Prevented (%)	–	21	35	61	64	76
RELLIS-3D + KITTI-360 (same runs)						
Runs	9	0	6	7	7	20
PME gate	0/9	–	2/6	5/7	6/7	19/20
OCAMO, z-gate	3/9	–	5/6	2/7	3/7	7/20
LTO, z-gate	4/9	–	5/6	5/7	4/7	12/20
OCAMO, own rule	0/9	–	0/6	0/7	0/7	0/20

(S-CSM). **(a) Appearance uncertainty.** Within both median Seg-Entropy strata, high-PME voxels flip (21.9%, 8.7%) and low-PME voxels almost never do (<0.1%), so PME separates vulnerability beyond appearance uncertainty. **(b) Components.** At σ=0.5°, PME selects voxels with a higher flip rate (39.3%) than its geometric-only and semantic-only parts. The geometric-only score uses the rotation block of Eq. (2). The semantic-only score is the ambiguity term of the PME decomposition, not the static Seg-Entropy of Table II. **Cost.** PME costs one segmentation pass (11.3 vs. 173.2 ms/frame for 16-pass MC Dropout, RTX 3080 Ti).

D. Detection of Map Damage

Alarms against damage. Table III sorts the fixed-offset runs by the mIoU loss of the final map and reports the first alarm, when updates would stop, and the share of new label errors that this stop prevents (0 for a miss). No offset-free replay alarms, and the more a run damages the map, the more often and the earlier the gate alarms. The gate detects 91% of the runs that need recalibration (over 5% mIoU loss), and four of its five misses come from one RELLIS-3D sequence (Sec. V). We sort by damage because 1° costs 11.1% on RELLIS-3D but 2.1% off-road.

Monitored sets. In Table V, mIoU↓ is the mean relative final-map loss (%), and the PME, ramp row counts alarms under the ramp. At 1°, the Random and Global sets detect

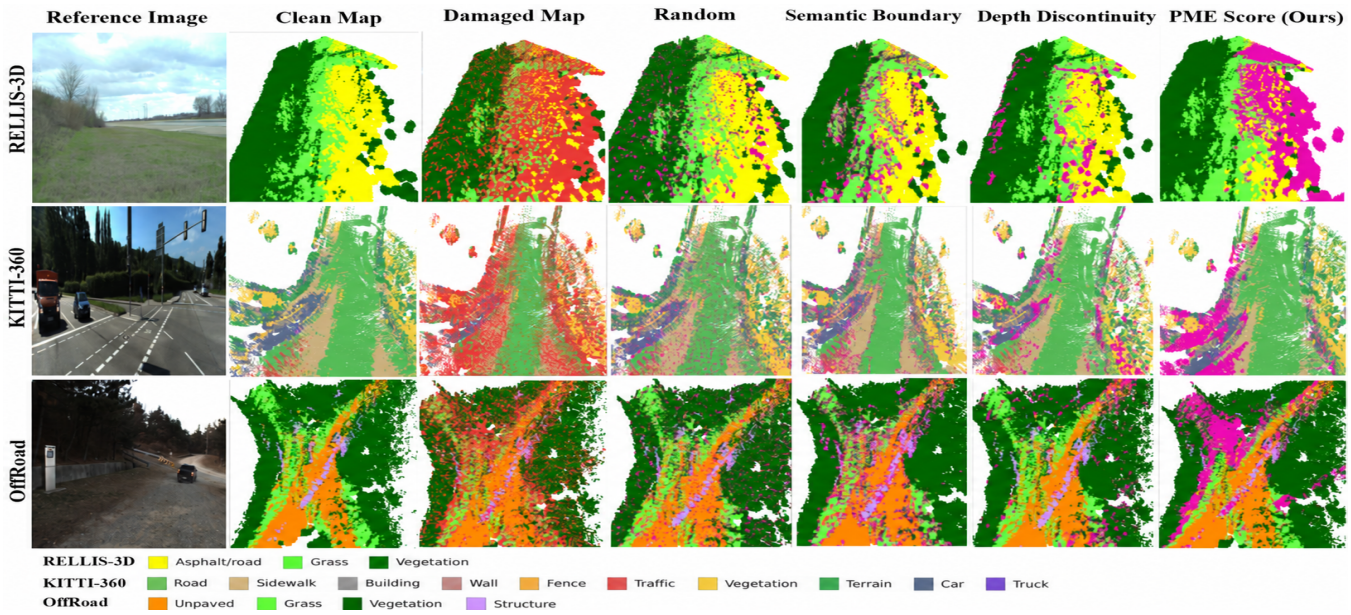


Fig. 5. Selection on RELLIS-3D, KITTI-360, and off-road maps (oblique views). Red marks voxels that one draw from the estimated 4° flip probability relabels. Magenta marks each rule's Top-10% (the monitored set is the Top-1%).

TABLE IV

ONLINE CALIBRATION MONITORS UNDER FIXED ROTATIONS θ, WITH RUN-LEVEL ALARMS AND CHECKPOINT AUROC (MEAN±STD). BOLD AND UNDERLINE MARK THE BEST AND SECOND-BEST PER DATASET IN THE WINDOWED AND AUROC COLUMNS.

				Runs						Checkpoint AUROC at θ				
				False alarm			Detected (1°)							
Dataset	Method	Decision level	Setup per route	Per-frame	Windowed	Native	Per-frame	Windowed	Native	0.5°	1°	2°	3°	ms
RELLIS-3D	OCAMO [4]	sensor pair	–	2/5	2/5	0/5	4/5	3/5	0/5	0.47±0.07	0.49±0.06	0.59±0.05	0.56±0.11	143–328
	LTO [4]	sensor pair	–	3/5	2/5	5/5	3/5	3/5	5/5	0.54±0.06	0.58±0.05	0.60±0.04	0.65±0.06	143–328
	Cal or No Cal?‡, other domain	sensor pair	none	3/5	**0/5**	5/5	4/5	0/5	5/5	0.50±0.02	0.48±0.02	0.47±0.03	0.51±0.03	26.5*
	Cal or No Cal?‡, same route	sensor pair	train 5.0–12.7 min	2/5	–	5/5	2/5	–	5/5	0.52±0.07	0.62±0.13	**0.88±0.08**	**0.99±0.01**	26.5*
	Ours (PME gate)	accum. map	clean pass 20–59 s	–	**0/5**	–	–	**4/5**	–	**0.59±0.21**	**0.78±0.27**	0.81±0.27	0.88±0.24	0.02–0.04
KITTI-360	OCAMO [4]	sensor pair	–	4/4	1/4	0/4	4/4	0/4	0/4	0.37±0.03	0.59±0.05	0.68±0.08	0.66±0.05	67–133
	LTO [4]	sensor pair	–	1/4	2/4	4/4	0/4	1/4	4/4	0.32±0.03	0.70±0.10	0.89±0.07	0.85±0.08	67–133
	Cal or No Cal?‡, same route	sensor pair	train 9.4–192 min	3/4	1/4	3/4	3/4	**4/4**	4/4	**0.73±0.03**	**0.95±0.04**	**1.00±0.00**	**1.00±0.01**	26.5*
	Ours (PME gate)	accum. map	clean pass 103 s (03)	–	**0/4**	–	–	**4/4**	–	0.51±0.06	0.59±0.09	0.96±0.03	**1.00±0.01**	0.12
Off-road	*Cal or No Cal?*‡, other domain	sensor pair	none	0/5	**0/5**	5/5	0/5	0/5	5/5	0.48±0.05	0.49±0.09	0.48±0.04	0.47±0.06	26.5*
	Cal or No Cal?‡, same route	sensor pair	train 0.5–3.3 min	3/5	–	4/5	3/5	–	4/5	**0.69±0.10**	**0.87±0.06**	**1.00±0.01**	0.99±0.02	26.5*
	Ours (PME gate)	accum. map	clean pass 3–17 s	–	**0/5**	–	–	**4/5**	–	0.57±0.07	0.73±0.22	**1.00±0.00**	**1.00±0.00**	0.04–0.07

5 and 4 of the 14 runs, against 12 for PME, so the signal lives in the monitored set. Seg-Entropy detects as often as PME but alarms on 1/5, 3/4, and 3/5 offset-free replays, and Boundary on 4/5 off-road ones.

Stronger frontend. A held-out SegFormer-B2 frontend detects fixed offsets of 0.5/1/1.5/2/3/4° in 2/2/4/4/4/5 RELLIS-3D runs, against 2/4/4/4/4/5 for MobileNetV3, and raises no alarm on five offset-free replays.

Unseen session and next-day re-drives. On next-day re-drives of the same route, the gate almost never alarms without an offset and detects offsets from 1° on two of the three drives, whereas a plain map comparison and a baseline from another route break down under day-to-day change. Table VI applies the frozen protocol to three re-drives of an off-road route that took no part in selection, training, or tuning, and Route is the share (%) of the first route that a re-drive covers. We select the monitored set once on the first drive and read it on each re-drive through GPS-matched registration, and the baseline comes from scans of the first drive that the selection does not use. Comparing the current map with the commissioning map, which also needs no labels or training, detects any offset trivially on the same recording but alarms on most offset-free checks across drives (Map diff). Its disagreement without an offset already reaches z=4.6–15.7 against the commissioning level, as registration error, lighting, and vegetation likely relabel more voxels than the offset. With the baseline of another sequence (a different route), 15/20, 12/20, and 3/12 pairs on RELLIS-3D, off-road, and KITTI-360 raise a false alarm. On urban KITTI-360, only 2.1% of the offset-free checkpoints alarm and detection from 1° largely holds, so the baseline partly transfers across urban routes, but a new off-road route needs re-commissioning.

Translation. In Table VII, Global is the share of voxels relabeled against the offset-free map, PME/Global the same share in the monitored set over it (random 1%, 1.0–1.1×), and Runs the alarming sequence–direction runs (offset-free, 0/5). 10–50 cm camera-frame offsets move projected points as much as 0.5°–2.5° rotations, and although PME perturbs translations only at 2 cm, its monitored set relabels 3.7–5.4×

TABLE V

DETECTIONS PER MONITORED SET AT FIXED ROTATIONS θ. BOLD AND UNDERLINE MARK THE TOP TWO PER DATASET AND COLUMN.

Dataset	Set	False alarm	0.5°	1°	1.5°	2°	3°	4°
RELLIS-3D	PME (ours)	**0/5**	**2/5**	**4/5**	**4/5**	4/5	4/5	**5/5**
	Seg-Entropy	1/5	**2/5**	**4/5**	**4/5**	**5/5**	**5/5**	**5/5**
	Boundary	**0/5**	1/5	**4/5**	**4/5**	4/5	4/5	4/5
	Random	**0/5**	1/5	1/5	3/5	3/5	3/5	4/5
	Global	**0/5**	0/5	1/5	3/5	3/5	3/5	3/5
	mIoU↓ (%)	0	5.5	11.1	16.8	22.2	31.0	37.2
	PME, ramp	–	2/5	2/5	3/5	4/5	4/5	5/5
KITTI-360 (4)	PME (ours)	**0/4**	2/4	**4/4**	**4/4**	**4/4**	**4/4**	**4/4**
	Seg-Entropy	3/4	2/4	**4/4**	**4/4**	**4/4**	**4/4**	**4/4**
	Boundary	3/4	**3/4**	3/4	**4/4**	**4/4**	**4/4**	**4/4**
	Random	1/4	2/4	2/4	2/4	2/4	2/4	3/4
	Global	1/4	2/4	1/4	2/4	2/4	2/4	2/4
	mIoU↓ (%)	0	2.6	8.4	14.4	19.6	28.2	34.9
	PME, ramp	–	1/4	4/4	4/4	4/4	4/4	4/4
Off-road	PME (ours)	**0/5**	1/5	**4/5**	**5/5**	**5/5**	**5/5**	**5/5**
	Seg-Entropy	3/5	2/5	**4/5**	**5/5**	**5/5**	**5/5**	**5/5**
	Boundary	4/5	**3/5**	3/5	**5/5**	**5/5**	**5/5**	**5/5**
	Random	1/5	1/5	2/5	2/5	2/5	2/5	3/5
	Global	2/5	2/5	2/5	2/5	2/5	3/5	3/5
	mIoU↓ (%)	0	0.8	2.1	3.8	5.7	9.5	13.5
	PME, ramp	–	1/5	4/5	5/5	5/5	5/5	5/5

TABLE VI

NEXT-DAY RE-DRIVES WITH THE MONITORED SET OF THE FIRST DRIVE. CLEAN AND MAP DIFF COUNT ALARMS ON OFFSET-FREE CHECKS.

Drive	Route	Clean	Map diff	0.5°	1°	1.5°	2°	3°
Re-drive A	74	1/67	41/67	–	✓	✓	✓	✓
Re-drive B	31	0/40	33/40	–	–	–	–	✓
Re-drive C	14	0/16	16/16	✓	✓	✓	✓	✓

as many voxels as the whole map. At similar whole-map relabeling, the gate detects translation less often than rotation (9/15 at 7.3% vs. 4/5 at 6.1%).

E. Why Online Calibration Alone Is Not Enough

We test four reasons, (i)–(iv), why online calibration alone does not protect the map, and then what the map receives until correction.

(i) Calibrator validity does not track map damage. Table IV compares monitors on fixed offsets under the per-frame, windowed, and native rules of Sec. IV-A. OCAMO's own rule ($V{>}0.5$ valid, σ_d from the other sequences, the released tracker bounds) flags no frame, so the index stays above 0.5 on all 32 RELLIS-3D and KITTI-360 runs at 1°–4° that lose over 5% mIoU, whereas the PME gate alarms on 29 of them. OCAMO updates this index every frame, so even continuous calibration misses these damaged maps. By damage (lower block of Table III, runs with OCAMO and LTO traces), their windowed z-gates alarm on 3/9 and 4/9 offset-free replays (PME 0/9) and exceed PME only in the 2–5% bin (5/6 vs. 2/6). *Cal or No Cal?* needs a GPU training run per route and, under its own rule, alarms on 12 of 14 offset-free replays. Its checkpoint AUROC exceeds ours at 1° on KITTI-360 and off-road and from 2° on RELLIS-3D, and with temporal smoothing it matches our gate on KITTI-360 (1/4 false alarms, 4/4 detections).

(ii) Faults a constant-extrinsic calibrator cannot repair. A camera–LiDAR time offset leaves T_{CL} correct, but under ego twist $\dot{\boldsymbol{\xi}}$ a lag of k frames acts as an extrinsic error $\delta\boldsymbol{\xi} \approx k\Delta t\,\dot{\boldsymbol{\xi}}$, to which Eq. (3) applies. This error varies with speed and turning, so a constant-T_{CL} calibrator absorbs only its mean. Shifting the image–scan pairing by $k{=}1$–3 frames (0.1 s each) loses 9.2–21.5% mIoU on RELLIS-3D, as much as 1°–2°, and the gate alarms in 2/5–3/5 runs. On KITTI-360 (mean of 03–06), 0.1 s and 0.3 s lose 14.0% and 27.4%, more than 1° and 2°, and the gate alarms on 57% and 96% of the checks. Applied every frame, the rotation-only OCAMO estimate removes 38%/22% of the 1°/2° RELLIS-3D mIoU loss but none of the time-offset damage.

A real fault. All our off-road data pair images and scans by timestamp. Pairing them by index instead shifts them by 137 ms on three of four raw off-road logs. On two long drives, the monitored set relabels about 9× as often as the map, and the gate alarms at 70–78% of the run.

(iii) Cost on a correct map and total compute. Table VIII compares periodic and alarm-triggered calls on KITTI-360, where a Calib-Anything call returns within the run. Loss is relative to the uncorrected offset-free map (8.9 and 20.6% without correction), Calib-Anything runs on 8 threads, and its periodic rows apply each estimate at once as a best case. Periodic calls re-estimate a still-correct extrinsic and cost 3.5–3.9% mIoU on the offset-free replay, where the gate makes no call. Under drift, the gate recovers comparably to periodic Calib-Anything, and it admits a calibrator too slow to run periodically. One Calib-Anything [9] call at the alarm, applied after its measured 223 s with held and replayed writes, leaves 4.7% loss at both 1° and 2°, against 7.4% and 16.8% for OCAMO every 10 s.

(iv) Repair depends on the calibrator and the domain.

TABLE VII

TRANSLATION FAULTS ON RELLIS-3D, THREE DIRECTIONS PER SEQUENCE.

Fault	Global (%)	PME/Global	Runs	First alarm (%)
Rotation 1°	6.1	5.7	4/5	26
Rotation 2°	11.1	4.6	4/5	25
Translation 10 cm	2.7	4.7	6/15	23
Translation 20 cm	5.1	4.4	8/15	20
Translation 30 cm	7.3	4.2	9/15	35
Translation 50 cm	11.4	3.7	12/15	23
1°+10 cm	6.7	5.4	9/15	26
2°+20 cm	11.9	4.3	13/15	26

TABLE VIII

CALIBRATOR AND TRIGGER ON KITTI-360 (TOP, 13–15 MIN RUNS) AND RELLIS-3D (BOTTOM), WITH FINAL MIOU LOSS (%) AND TIME PER CALL.

Calibrator	Trigger	Call	Clean	1°	2°
OCAMO	every 10 s	7.5 s	3.5	7.4	16.8
OCAMO	once per alarm	7.5 s	0	9.8	18.7
Calib-Anything	every 20 s	223 s	3.9	4.5	4.6
Calib-Anything	alarm, hold protected	223 s	0	5.0	6.2
Calib-Anything	alarm, hold all	223 s	0	4.7	4.7
OCAMO	every 10 s	20 s	4.8	6.4	13.8
OCAMO	once per alarm	20 s	0	10.5	19.5
LCCNet	every 10 s	0.2 s	2.9	5.6	13.8
LCCNet	once per alarm	0.2 s	0	9.0	17.0
Check	alarm, hold protected	60 s	0	4.0	9.8

TABLE IX

FINAL GLOBAL MIOU (%) UNDER THE 0°–3° RAMP.

Dataset	Detected	Clean	Ungated	Corrected	Delayed	Recovery
RELLIS-3D	4/5	36.40±3.85	25.81±2.94	34.44±3.17	32.54±2.97	79.4±13.3
KITTI-360	4/4	54.12±2.88	39.66±2.87	53.59±3.38	47.13±3.13	96.3±3.8
Off-road	5/5	40.13±4.10	36.54±3.46	39.05±3.87	37.75±3.76	74.1±22.5

On RELLIS-3D, we commission the extrinsic once with OCAMO (0.4°–0.8° from the provided one) and inject faults on top of it (9.1 and 20.1% without correction). Periodic OCAMO and LCCNet (KITTI weights, finest stage, as all five stages diverge) again cost 4.8 and 2.9% on the correct map but remove only 30–38% of the drift loss. OCAMO every 10 s overlaps its 20 s calls and applies each estimate at once, a best case. After a 1° offset, one call leaves a median rotation error of 1.6° for OCAMO (1.8° with 3° bounds per axis), 0.9° for LCCNet, and 5.8°–6.2° for Calib-Anything on three of four sequences. One call at the alarm thus loses more than periodic calls, as on KITTI-360, because its error persists while periodic calls replace theirs. The gate decides when to check and what to hold, not how to repair, so a check that restores the commissioned extrinsic 60 s after the alarm, with the hold, leaves 4.0 and 9.8% (Check row), below every periodic calibrator and with no cost on the correct map.

What the map receives until the check. On RELLIS-3D, holding all writes from the first alarm avoids 79% of the corrupted voxels at the run's end but leaves only 20–47% of the map observed if the check comes 20 s or more later. We therefore hold only the protected set, the top-r voxels by commissioning PME, and assume that a check D later removes the cause and replays the buffer. We call a voxel corrupted if it differs from the offset-free map at its last write. With D=60 s, holding the protected set with r=20% avoids 60% and 53% of the corruption on RELLIS-3D and KITTI-360 03–06 (16 alarmed rotation and time-offset runs each), against 20% for random and 46–50% for boundary proximity. After replay, the mIoU drop falls from 16.2% to 6.1% on RELLIS-3D and from 21.6% to 10.9% on KITTI-360.

Map quality after an alarm-triggered correction. On the 3° ramp, each replay switches to the clean extrinsic either at the first alarm (Corrected) or halfway through the rest of the run (Delayed), and neither fixes the voxels that the replay writes before the switch. Over the detected runs (Table IX), the Corrected replay recovers 79.4, 96.3, and 74.1% of the lost mIoU on average (worst 60.3, 89.7, and 35.0%) on RELLIS-3D, KITTI-360, and off-road.

V. CONCLUSIONS

Camera–LiDAR misalignment concentrates label errors in the few voxels whose projections cross label changes (LFA), which calibration-level methods do not observe. PME finds these voxels from one clean traversal of a repeated route without training, and a label-free gate on them raises almost no alarm on offset-free replays and detects 52 of 57 runs that lose over 5% mIoU as well as time offsets. Holding the protected set, 20% of the voxels, until a check avoids 53–60% of the corruption. The gate therefore sits above online calibration rather than replacing it, deciding when to call a calibrator and which writes to hold until the correction. Independent of calibrator accuracy, its alarm spares a correct map from periodic calls, admits too-slow calibrators, and still protects the map on RELLIS-3D, where no tested calibrator restores it.

Limitations. The gate misses RELLIS-3D 00002 below 4°, also with the SegFormer frontend. There, 84% of the monitored set lies inside paved road and only 7% on class boundaries, against 41–65% on the other sequences. Its map entropy varies so much (σ_h=0.18 vs. 0.01–0.11) that a 1° offset shifts it by only $0.28\,\sigma_h$. At 0.5°, it alarms in only 5 of 14 runs. A new route needs re-commissioning, and the baseline only partly transfers across urban routes (Sec. IV-D). Apart from the pairing fault (Sec. IV-E), we test only injected faults, as in [3], [5], [10].

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